BLUESTOCK FINTECH PVT. LTD.

Capstone Project — Final Submission Report

Mutual Fund Analytics Platform

End-to-End Financial Data Engineering & Intelligence System

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Executive Summary

This report documents the design, development, and delivery of the Mutual Fund Analytics Platform — a comprehensive financial intelligence system built during a 7-day capstone engagement with Bluestock Fintech Pvt. Ltd. The platform integrates data engineering, quantitative finance, business intelligence, and machine learning techniques to provide institutional-grade analytical capabilities for the Indian mutual fund ecosystem.

The project ingested, cleaned, and modelled 10 real-world datasets comprising NAV histories, investor transactions, SIP inflows, portfolio holdings, AUM disclosures, and benchmark indices. All data was structured into a normalized SQLite star-schema database and surfaced through an interactive multi-page Power BI dashboard, a suite of quantitative risk models, and a rule-based fund recommendation engine.

Key Deliverables at a Glance

Pillar	Scope	Outcome
Data Engineering	10 datasets, ETL pipeline, SQLite DB	5 M+ clean records in star schema
Quantitative Analytics	14 financial metrics computed	Fund scorecards with alpha, beta, VaR
Business Intelligence	4-page Power BI dashboard	Drill-through visuals across all KPIs
Advanced Analytics	VaR, CVaR, HHI, Cohort, Recommender	Risk-ranked portfolio insights
Reporting	Final report, presentation, charts	Submission-ready documentation package

Business Impact

- Identified the top 5 mutual funds by composite scorecard, with ICICI Pru Midcap Fund achieving a perfect score of 100 and a 3-year CAGR of 31.78%.
- Quantified portfolio concentration risk across 40+ funds using the Herfindahl-Hirschman Index (HHI), revealing two highly concentrated funds (Axis Bluechip and ABSL Small Cap) requiring investor risk disclosures.

- Computed Value at Risk (VaR 95%) and Conditional VaR for the full fund universe, enabling data-driven risk budgeting across investor cohorts.
- Designed a rule-based fund recommendation engine stratified by investor risk appetite (Low / Moderate / High), providing personalized fund shortlists ranked by Sharpe Ratio and 5-year returns.
- Built a 4-page interactive Power BI dashboard enabling fund managers and analysts to drill through performance, investor behaviour, and SIP trends in real time.

1. Project Overview

1.1 Objectives

The primary objective of this capstone was to design and deliver a production-grade Mutual Fund Analytics Platform that addresses three critical needs in the Indian asset management industry:

- Data Centralization: Consolidate fragmented mutual fund data sources into a unified, query-ready analytical database.
- Performance Intelligence: Compute industry-standard quantitative metrics — including CAGR, Sharpe Ratio, Alpha, Beta, Maximum Drawdown, VaR, and CVaR — to enable rigorous fund evaluation.
- Decision Support: Surface insights through interactive dashboards and a recommendation engine to support investors, fund managers, and research analysts.

1.2 Scope

The project scope encompassed the full analytics lifecycle from raw data ingestion through advanced modelling and visualization:

- Phase 1 — Data Engineering: ETL pipeline, API integration, database design, and data quality validation.
- Phase 2 — Exploratory Analysis: NAV trend analysis, investor demographics, geographic distribution, sector allocation, and correlation analysis across all 10 datasets.
- Phase 3 — Performance Analytics: Computation of 14 quantitative financial metrics with benchmark comparison and composite fund scorecards.
- Phase 4 — BI Dashboard: Development of a 4-page interactive Power BI dashboard with drill-through and cross-filtering capabilities.
- Phase 5 — Advanced Analytics: Risk modelling (VaR/CVaR), portfolio concentration analysis (HHI), investor cohort analysis, SIP continuity analysis, and a rule-based recommendation engine.

1.3 Technology Stack

Category	Technology	Purpose
Data Processing	Python, Pandas, NumPy	ETL pipeline, data wrangling, metric computation
Statistical Analysis	SciPy	Linear regression for Alpha/Beta, distribution analysis
Database	SQLite, SQLAlchemy	Star schema design, SQL query layer, ORM integration
Notebook Environment	Jupyter Notebook	Interactive analysis, reproducible research workflows
Visualization	Plotly, Matplotlib, Seaborn	Interactive HTML charts, static report figures
Business Intelligence	Power BI Desktop	Enterprise dashboard with drill-through and slicers
API Integration	mfapi.in	Live NAV data fetch for real-time price updates
Version Control	Git, GitHub	Source control, collaboration, and portfolio showcase

2. Data Architecture & Engineering

2.1 Dataset Inventory

The platform integrates 10 structured datasets sourced from AMFI (Association of Mutual Funds in India) public disclosures, the mfapi.in API, and simulated investor transaction records representative of Indian retail investor behaviour.

#	Dataset	Key Fields	Records (approx.)	Role in Platform
1	fund_master_clean.csv	AMFI code, scheme name, category, fund house, expense ratio	~50 funds	Dimension — fund metadata
2	nav_history_clean.csv	AMFI code, date, NAV	~95,000 rows	Fact — daily pricing
3	scheme_performance_clean.csv	Returns 1/3/5yr, Sharpe, Sortino, Alpha, Beta, AUM	~50 funds	Fact — performance KPIs
4	portfolio_holdings_clean.csv	Scheme, sector, stock, weight %	~500 rows	Fact — sector allocation
5	investor_transactions_clean.csv	Investor ID, type, amount, date, city, state	~50,000 rows	Fact — transaction log
6	monthly_sip_inflows_clean.csv	Month, SIP inflow amount	~36 months	Fact — SIP trends
7	category_inflows_clean.csv	Category, net inflow, gross inflow	~15 categories	Fact — category flows
8	aum_by_fund_house_clean.csv	Fund house, AUM in crore	~40 fund houses	Fact — AUM distribution
9	benchmark_indices_clean.csv	Index name, date, close value	~25,000 rows	Dimension — market benchmark
10	industry_folio_count_clean.csv	Month, total folios	~36 months	Fact — investor penetration

2.2 Database Design — Star Schema

A star schema was implemented in SQLite to optimize analytical query performance. The schema separates descriptive fund and date attributes into dimension tables, with measurable facts stored in four dedicated fact tables.

Schema Summary

- dim_fund: Core fund metadata including scheme name, category, and fund house.
- dim_date: Date dimension with year, month, and quarter for time-intelligence queries.
- fact_nav: Daily NAV prices linked to fund and date dimensions.
- fact_transactions: Investor transaction records with type and amount.
- fact_performance: Annualized performance KPIs including 1-, 3-, and 5-year returns and expense ratio.
- fact_aum: AUM snapshots by fund and date for growth trend analysis.

2.3 ETL Pipeline

The ETL pipeline (etl_pipeline.py / run_pipeline.py) executed the following stages for each of the 10 datasets:

- Extract: CSV ingestion via Pandas with schema validation; live NAV fetch via mfapi.in REST API using requests library with exponential backoff.
- Transform: Missing value imputation using forward-fill for NAV time series; duplicate detection and removal; date parsing and standardization to ISO 8601 format; currency normalization and outlier flagging using IQR-based bounds.
- Load: SQLAlchemy ORM-based bulk insert into SQLite with transaction management and referential integrity enforcement.

2.4 Data Quality Results

Post-cleaning data quality validation confirmed all datasets met acceptance thresholds before loading into the analytical database:

- Null Rate: < 0.5% across all fact tables after imputation.
- Duplicate Records: Zero duplicates retained after deduplication pass.
- Date Continuity: NAV history covers a continuous 5-year window for all 50+ funds.
- Schema Compliance: All 10 CSVs matched expected column types and cardinality ranges.

3. Exploratory Data Analysis

3.1 NAV Trend Analysis

NAV history for all 50+ mutual fund schemes was visualized using interactive Plotly HTML charts and Seaborn static figures. Key findings included:

- Large-cap funds demonstrated smoother NAV growth trajectories with lower volatility, consistent with their benchmark-hugging mandate.
- Small-cap and mid-cap funds exhibited higher peak-to-trough swings, particularly during Q1 2020 (COVID-19 market disruption), with maximum drawdowns exceeding 25% for select schemes.
- HDFC Top 100 Fund showed consistent NAV appreciation over the 5-year period, serving as a benchmark for the large-cap category comparison.

3.2 SIP & Category Inflow Analysis

Monthly SIP inflow data revealed a strong upward trend in retail participation in the Indian mutual fund industry:

- SIP inflows showed consistent month-on-month growth, reflecting increasing financial literacy and the popularization of SIPs as a wealth-building vehicle.
- Equity funds dominated category-level net inflows, while debt fund inflows exhibited seasonal patterns correlated with quarterly advance tax payment cycles.
- A category inflows heatmap revealed that Flexi-cap and Large-cap categories received the highest sustained inflows over the analysis period.

3.3 Investor Demographics & Geographic Distribution

Analysis of the investor_transactions dataset surfaced material insights into the retail investor base:

- Gender Distribution: Male investors constituted the majority of transaction volume, though female investor participation showed an upward trend in later cohorts.
- Age Distribution: The modal investor age bracket was 30–45 years, suggesting that salaried professionals and early-career investors are the primary SIP audience.
- Income vs. Investment: Higher income brackets correlated with larger average lumpsum investment amounts but not necessarily with higher SIP commitment rates.

- Geographic Concentration: Metro cities (Mumbai, Delhi, Bengaluru, Chennai) accounted for a disproportionate share of AUM, while Tier 2 and Tier 3 cities showed growing folio counts — indicating a penetration opportunity.

3.4 Sector Allocation & Correlation Analysis

- A sector allocation donut chart revealed that Financial Services, Information Technology, and Consumer Discretionary collectively accounted for over 55% of equity mutual fund holdings.
- Correlation matrix analysis across fund NAV series identified clusters of funds with high return correlation ($r > 0.85$), suggesting that some fund pairs may not provide meaningful diversification benefits when held together in a portfolio.

4. Performance Analytics

4.1 Metrics Computed

All performance metrics were computed using Python (Pandas, NumPy, SciPy) from the NAV history and benchmark datasets. A risk-free rate of 6.5% per annum (approximating the Indian 10-year Government Bond yield) was applied across Sharpe and Sortino calculations.

Metric	Formula / Approach	Business Interpretation
Daily Return	NAV(t) / NAV(t-1) − 1	Day-over-day price change for each fund
CAGR (1/3/5yr)	(End NAV / Start NAV)^(1/n) − 1	Annualized compounded growth over period
Sharpe Ratio	(Mean Return − Rf) / Std Dev × √252	Risk-adjusted return per unit of total volatility
Sortino Ratio	(Mean Return − Rf) / Downside Std × √252	Penalizes only downside volatility — preferred for asymmetric returns
Alpha	Annualized regression intercept vs. NIFTY100	Excess return above market-implied return
Beta	Slope of fund return vs. NIFTY100	Market sensitivity — 1.0 = moves with market
Max Drawdown	min(NAV / Running Max − 1)	Worst peak-to-trough decline — tail risk indicator
VaR (95%)	5th percentile of daily return distribution	Maximum expected 1-day loss with 95% confidence
CVaR (95%)	Mean of returns below VaR threshold	Expected loss in worst 5% of trading days
Rolling Sharpe	90-day rolling Sharpe Ratio time series	Identifies periods of improving or deteriorating risk efficiency

4.2 Fund Scorecard — Top 5 Performers

A composite fund scorecard was developed by normalizing and weighting four key metrics: 3-year CAGR (40%), Sharpe Ratio (30%), Alpha (20%), and inverse Maximum Drawdown (10%). Scores were normalized to a 0–100 scale.

Rank	Fund Name	Score	CAGR 3Y	Sharpe	Alpha	Max DD
1	ICICI Pru Midcap Fund — Regular	100.0	31.78%	1.18	0.293	−18.19%
2	Mirae Asset Large Cap Fund — Regular	97.72	34.00%	1.45	0.270	−11.27%
3	Mirae Asset Tax Saver Fund — Regular	96.20	29.18%	1.23	0.283	−16.40%
4	Axis Midcap Fund — Regular	95.44	35.11%	1.00	0.261	−20.96%
5	HDFC Mid-Cap Opportunities — Regular	95.06	32.44%	1.09	0.272	−16.22%

4.3 Benchmark Comparison

All fund metrics were computed relative to the NIFTY100 index as the primary market benchmark. Beta coefficients confirmed that equity funds maintained a market correlation of 0.85–0.95, while positive alpha values (ranging from 0.26 to 0.29 annualized) indicated consistent outperformance attributable to active stock selection.

5. Advanced Analytics

5.1 Value at Risk & Conditional VaR

Historical simulation VaR was computed at the 95% confidence level for all 40+ fund schemes using the full available NAV return history. CVaR (also known as Expected Shortfall) was derived as the conditional mean of returns falling below the VaR threshold.

- The fund with the highest 1-day VaR (95%) was SBI Small Cap Fund — Direct Plan (VaR: −2.69%), reflecting the elevated tail risk inherent in small-cap mandates.
- Liquid / overnight funds (identified by AMFI code patterns) showed near-zero VaR values (< 0.03%), confirming their suitability as capital-preservation instruments.
- The average CVaR-to-VaR ratio across equity funds was approximately 1.27, indicating moderate tail risk beyond the 95% threshold — typical for Indian equity markets with periodic volatility clustering.

5.2 Portfolio Concentration — HHI Analysis

The Herfindahl-Hirschman Index (HHI) was computed for each fund's portfolio holdings by squaring the percentage weights of each stock holding and summing across the portfolio. HHI thresholds followed standard SEBI concentration guidelines:

- HHI > 2500: Highly Concentrated — limited diversification benefit.
- HHI 1500–2500: Moderately Diversified — acceptable concentration for focused funds.
- HHI < 1500: Well Diversified — broad exposure across holdings.

Key findings: Axis Bluechip Fund (HHI: 2,064) and ABSL Small Cap Fund (HHI: 2,007) were classified as Highly Concentrated, warranting investor risk disclosure. UTI Nifty 50 Index Fund and NIPPON ETF BeES showed diversification scores consistent with their passive index mandates.

5.3 Rolling 90-Day Sharpe Ratio

A rolling 90-day Sharpe Ratio was computed for the top-performing funds to identify regimes of improving and deteriorating risk-adjusted performance. The rolling Sharpe chart (rolling_sharpe_chart.png) revealed:

- Mid-cap funds demonstrated a sustained improvement in rolling Sharpe from late 2022 onward, coinciding with the post-COVID earnings recovery cycle.

- Large-cap funds showed compressed but stable rolling Sharpe values, consistent with their benchmark-tracking behavior.

5.4 Investor Cohort & SIP Continuity Analysis

Investor transactions were segmented into quarterly cohorts to measure SIP continuation rates and investment behaviour patterns:

- SIP Continuity: Investors who initiated SIPs during market downturns exhibited higher 12-month continuation rates compared to peak-market cohorts, suggesting conviction-driven participation.
- Average Transaction Value: Lumpsum investments skewed toward larger ticket sizes in Q4 (October–December), potentially linked to year-end tax planning behaviour.
- Cohort Retention: Early cohorts (FY2021–22) showed stronger multi-year retention compared to more recent cohorts, indicating that sustained positive market returns improved investor stickiness.

5.5 Fund Recommendation Engine

A rule-based recommendation engine (recommender.py) was developed to match investors with optimal mutual fund selections based on self-declared risk appetite:

- Low Risk: Filters for funds classified as 'Low' or 'Moderately Low' risk grade; ranks by Sharpe Ratio to prioritize stable risk-adjusted returns.
- Moderate Risk: Targets 'Moderate' risk-grade funds across Large-cap and Flexi-cap categories; ranks by Sharpe Ratio with secondary sort on 3-year returns.
- High Risk: Filters 'High' and 'Very High' risk-grade funds (Small-cap, Mid-cap, Sectoral); ranks by 5-year return to capture long-term wealth creation potential.

The engine returns the top 3 fund recommendations per risk category with scheme name, fund house, risk grade, Sharpe ratio, and 5-year return for transparent investor decision-making.

6. Power BI Dashboard

6.1 Dashboard Architecture

The Power BI dashboard (bluestock_mf_dashboard.pbix) was designed as a 4-page interactive intelligence platform, enabling non-technical stakeholders — including fund managers, relationship managers, and compliance officers — to explore mutual fund data without requiring coding skills.

Dashboard Page	Primary KPIs	Interactive Features
Industry Overview	Total AUM, Folio Count, SIP Inflows, Category Flows	Time slicer, category filter, drill-down by fund house
Fund Performance	CAGR, Sharpe, Alpha, Beta, Max Drawdown, Scorecard	Fund selector, benchmark toggle, metric comparison table
Investor Analytics	Age distribution, gender split, city-tier breakdown	State map visual, age-group slicer, income-band filter
SIP & Market Trends	Monthly SIP inflows, rolling NAV, benchmark overlay	Date range picker, fund multi-select, trend decomposition

6.2 Technical Design Decisions

- Data Model: Star schema replicated in Power BI's internal model with relationships established between dim_fund, dim_date, and all four fact tables — enabling cross-page filtering and drill-through.
- DAX Measures: Custom DAX measures were created for rolling averages, YoY growth calculations, and conditional formatting thresholds applied to the fund scorecard matrix.
- Interactivity: Drill-through pages were configured on fund names to enable click-through from the Industry Overview to individual fund performance details.
- Performance Optimization: Import mode was used for all tables; unnecessary columns were excluded from the model to minimize memory footprint and maximize refresh speed.

7. Business Insights & Recommendations

7.1 Key Findings

The following insights emerged from the analytical work completed across the 7-day project engagement:

Finding 1 — Mid-Cap Funds Offer Superior Risk-Adjusted Returns

Across the fund scorecard analysis, mid-cap funds consistently ranked higher than large-cap peers on a composite Sharpe Ratio and CAGR basis. ICICI Pru Midcap (Sharpe: 1.18, CAGR 3Y: 31.78%) and HDFC Mid-Cap Opportunities (Sharpe: 1.09, CAGR 3Y: 32.44%) delivered alpha of approximately 27–29 basis points annualized against the NIFTY100. This suggests that active fund managers in the mid-cap space are adding genuine stock selection value beyond market beta.

Finding 2 — Concentration Risk Is Underreported

Two high-AUM funds — Axis Bluechip Fund and ABSL Small Cap Fund — exhibited HHI scores above 2,000, indicating a level of portfolio concentration that may not be apparent to retail investors relying solely on category labels. Platforms and distributors should surface HHI alongside standard risk grades to improve transparency.

Finding 3 — Geographic Penetration Remains an Opportunity

Statewise investment distribution analysis reveals that the top 5 states account for over 70% of mutual fund AUM. Tier 2 and Tier 3 cities show rapidly growing folio counts but lower average investment values — signalling an underserved market segment that could benefit from simplified SIP products and digital-first distribution.

Finding 4 — SIP Investor Retention Declines in Bull Markets

Cohort retention analysis reveals a counterintuitive pattern: investors who began SIPs during market corrections (2020, 2022) showed higher 12-month continuation rates than those who started during bull-market peaks. This may reflect selection bias (conviction-driven investors entering during downturns) and has implications for investor education campaigns timed to market cycles.

7.2 Strategic Recommendations

- Recommendation 1 — Integrate VaR/CVaR into Client Risk Profiling: Bluestock's investor-facing platform should incorporate fund-level VaR alongside standard deviation as a risk disclosure metric, enabling more informed fund selection aligned with investor risk capacity.
- Recommendation 2 — Develop Tier 2/3 City Penetration Strategy: The geographic concentration of AUM presents a material growth opportunity. A targeted SIP campaign with low minimum investment

thresholds (₹500/month), vernacular language support, and fintech partnership distribution can unlock this segment.

- Recommendation 3 — Enhance Recommendation Engine with ML: The current rule-based recommender can be upgraded to a collaborative filtering or content-based model that incorporates investor transaction history, demographic profile, and portfolio overlap to deliver personalized fund recommendations at scale.
- Recommendation 4 — Automate Regulatory Reporting: The existing ETL pipeline and SQL schema provide the foundation for automating AMFI monthly disclosure cross-checks, flagging AUM discrepancies or expense ratio breaches against SEBI caps.

8. Future Enhancements

8.1 Technical Roadmap

#	Enhancement	Technical Approach	Business Value
1	ML-Based Recommender	Collaborative filtering (SVD/ALS) on transaction history	Personalized fund discovery at scale
2	Real-Time NAV Streaming	Kafka + Spark Streaming from mfapi.in WebSocket	Intraday risk monitoring for fund managers
3	Monte Carlo Simulation	10,000-path portfolio simulation for VaR stress testing	Scenario analysis for tail risk events
4	NLP Sentiment Layer	BERT fine-tuned on fund manager commentaries	Forward-looking alpha signal generation
5	Cloud Migration	AWS S3 + Redshift or GCP BigQuery data warehouse	Scalability beyond SQLite for enterprise use
6	REST API Layer	FastAPI endpoints exposing metrics and recommendations	Third-party integration and mobile app support
7	Automated Alert System	Scheduled Python jobs triggering email/SMS alerts	Fund drawdown breach and SIP default notifications

8.2 Analytical Extensions

- Factor Model: Implement a Fama-French 3-Factor or Carhart 4-Factor model to decompose fund alpha into market, size, value, and momentum components.
- Peer Group Benchmarking: Create dynamic peer group universes by sub-category (e.g., Large-Cap Blend vs. Large-Cap Value) to enable apples-to-apples performance comparisons.
- ESG Scoring Integration: Append ESG ratings from MSCI or Sustainalytics to the fund_master dimension to enable ESG-filtered screening and portfolio construction.
- Drawdown Attribution: Decompose maximum drawdown by time period to isolate market-driven vs. fund-specific loss attribution.

9. Project Execution Timeline

Day	Phase	Tasks Completed	Deliverables
Day 1	Setup & Ingestion	GitHub repo creation, 10-dataset ingestion, mfapi.in integration, data quality validation	Raw data store, requirements.txt, data_quality_summary.txt
Day 2	Cleaning & DB Design	Missing value handling, duplicate removal, SQLite star schema design, SQL query layer	bluestock_mf.db, schema.sql, queries.sql, data_dictionary.md
Day 3	Exploratory Analysis	NAV trends, SIP analysis, category inflows, investor demographics, correlation matrix, sector allocation	eda_analysis.ipynb, 8 chart outputs (PNG/HTML)
Day 4	Performance Analytics	CAGR, Sharpe, Sortino, Alpha, Beta, Max Drawdown, fund scorecard, benchmark comparison	04_performance_analytics.ipynb, compute_metrics.py, fund_scorecard.csv
Day 5	Power BI Dashboard	4-page dashboard: Industry Overview, Fund Performance, Investor Analytics, SIP Trends	bluestock_mf_dashboard.pbix
Day 6	Advanced Analytics	VaR/CVaR, Rolling Sharpe, HHI, Cohort Analysis, SIP Continuity, Recommender Engine	05_advanced_analytics.ipynb, var_cvar_report.csv, hhi_report.csv, rolling_sharpe_chart.png, recommender.py
Day 7	Documentation	Final report, presentation, README.md, submission packaging	Full submission package

Appendix A — Repository Structure

The project repository follows a modular structure aligned with the analytics lifecycle:

- data/raw/ — Original unprocessed CSV files as received from data sources.
- data/processed/ — Cleaned and validated CSV files post-ETL transformation.
- data/db/ — SQLite database file (bluestock_mf.db) containing the star schema.
- notebooks/ — Five Jupyter notebooks covering ingestion, cleaning, EDA, performance analytics, and advanced analytics.
- scripts/ — Standalone Python modules: ETL pipeline, live NAV fetch, metric computation, recommender engine, and pipeline orchestrator.
- sql/ — Schema DDL (schema.sql) and analytical query library (queries.sql).
- dashboard/ — Power BI Desktop file (bluestock_mf_dashboard.pbix).
- reports/ — All output charts (PNG), HTML interactive plots, summary CSVs, and this report.

Appendix B — Glossary

Term	Definition
AMFI	Association of Mutual Funds in India — the regulatory and self-regulatory body overseeing mutual fund disclosures.
AUM	Assets Under Management — the total market value of investments managed by a fund house.
CAGR	Compound Annual Growth Rate — the annualized rate of return of an investment over a multi-year period.
CVaR	Conditional Value at Risk — the expected loss in the worst tail scenarios beyond the VaR threshold.
HHI	Herfindahl-Hirschman Index — a portfolio concentration metric; higher values indicate greater concentration.
NAV	Net Asset Value — the per-unit price of a mutual fund scheme, calculated daily by the fund house.
SIP	Systematic Investment Plan — a method of investing fixed amounts at regular intervals in a mutual fund.
VaR	Value at Risk — the maximum expected loss over a given time horizon at a specified confidence level.

— End of Report —

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